

# Voter Survey & Management System

Complete Role-Based Access Control (RBAC), UI Wireframes & Developer Technical Guide

This document serves as the official specification for building the multi-tier Voter Management, Field Survey & Progress Tracking System.

It details the form controls, cascading access scopes, validation rules, and screen layouts for **Super Admin (A1)**, **Second Level Admin (A2)**, and **Field Agent (A3)**.

TARGET ROLES

Super Admin (A1) • Sub Admin (A2) • Field Agent (A3)

MODULES INCLUDED

Authentication, User Registration, Master Data, Survey Engine, Dashboard

DOCUMENT VERSION

v1.0 (Developer Hand-off)

ARCHITECTURE STACK

Web Portal (React/Next) & Mobile Field App (PWA/Flutter) + REST API + PostgreSQL

# 1. System Overview & Permission Matrix

The system is structured as a 3-tier hierarchical portal for electoral verification, voter field surveys, and live analytics.

## 1.1 Role Definitions & Permission Matrix

| ROLE IDENTIFIER | ROLE TITLE                      | JURISDICTION SCOPE                               | PERMISSIONS & CAPABILITIES                                                                                                                                                                                                                                                           |
|-----------------|---------------------------------|--------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| A1              | Super Admin                     | Global (All Constituencies / Panchayats / Wards) | <ul style="list-style-type: none"><li>Full system access with NO restrictions.</li><li>Create &amp; manage Level 2 Admins (A2) and Agents (A3).</li><li>Manage Master Data (Caste, Job, Political Party).</li><li>View Global Panchayat-wise survey completion dashboards.</li></ul> |
| A2              | Second Level Admin (Supervisor) | Scoped (Assigned Panchayats & Wards only)        | <ul style="list-style-type: none"><li>Access strictly restricted to assigned geographic boundaries.</li><li>Can create Field Agents (A3) only within assigned scope.</li><li>View progress dashboards for assigned Panchayats &amp; Agents.</li></ul>                                |
| A3              | Field Agent (Surveyor)          | Scoped (Assigned Wards / Booths)                 | <ul style="list-style-type: none"><li>Mobile field survey access.</li><li>Search voter by EPIC No, Name, or Door Number.</li><li>View pre-filled voter data (Read-Only) and correct names.</li><li>Collect: Phone Number, Caste, Job, Party Affiliation.</li></ul>                   |

**Core Security Rule:** When registering any user (A1, A2, A3), entering a valid EPIC ID matching the VOTERS\_MASTER database is mandatory before activating the account.

## 1.2 Field Access & Data Collection Rules

| FIELD NAME            | DATA TYPE / CONTROL     | PRE-FILLED? | AGENT (A3) PERMISSION | SOURCE / RULE                                  |
|-----------------------|-------------------------|-------------|-----------------------|------------------------------------------------|
| EPIC Number           | Text (Disabled)         | YES         | LOCKED / READ-ONLY    | Unique Electoral ID from Database              |
| Panchayat / Ward      | Text (Disabled)         | YES         | LOCKED / READ-ONLY    | Geographic boundary from Database              |
| Door No / Address     | Text (Disabled)         | YES         | LOCKED / READ-ONLY    | Official registered address from DB            |
| Age & Sex             | Text / Radio (Disabled) | YES         | LOCKED / READ-ONLY    | Pre-filled from electoral list                 |
| First Name            | Text Input              | YES         | EDITABLE              | Allows field correction of spelling typos      |
| Last / Father Name    | Text Input              | YES         | EDITABLE              | Allows field correction of father/husband name |
| Phone Number          | 10-digit Tel Input      | NO          | REQUIRED ENTRY        | Regex: ^[6-9]\d{9}\$                           |
| Caste                 | Dropdown Selector       | NO          | REQUIRED ENTRY        | Dynamic options from <b>A1 Caste Master</b>    |
| Job / Occupation      | Dropdown Selector       | NO          | REQUIRED ENTRY        | Dynamic options from <b>A1 Job Master</b>      |
| Belongs to Party/Part | Dropdown Selector       | NO          | REQUIRED ENTRY        | Dynamic options from <b>A1 Party Master</b>    |

## 2. Screen Wireframes: Authentication & User Registration

### 2.1 Screen 1: Universal Login Screen (/login)

AUTH • UNIVERSAL LOGIN

Route: /login

Voter Survey Portal

Sign in with registered mobile & password

USER MOBILE NUMBER \*

+91 9876543210

PASSWORD \*

.....

LOGIN TO SYSTEM

### 2.2 Screen 2: User Registration Form (A1 & A2)

USER MANAGEMENT • CREATE USER ACCOUNT

Route: /admin/users/create

User Details & Cascading Jurisdiction Assignment

TARGET ROLE \*

Second Level Admin (A2)

Field Agent (A3)

\*(A2 Admins can only register A3 Agents)\*

ACCOUNT STATUS \*

Active (Yes)

Disabled (No)

USER MOBILE NUMBER \*

9840123456

PASSWORD \*

.....

EPIC ID (VOTER VERIFICATION) \*

TN/04/023/123456

Verify EPIC

✓ Verified Voter: R. Rajesh Kumar (Melur Panchayat, Ward 04)

JURISDICTION ASSIGNMENT (CASCAADING MULTI-SELECT)

CONSTITUENCY NUMBER / NAME (MULTI-SELECT) \*

124 - Madurai North

125 - Madurai South

+ Add more

PANCHAYAT (MULTI-SELECT - FILTERED BY CONSTITUENCY) \*

Melur

Alanganallur

Othakadai

+ Add more

WARD / S.NO (MULTI-SELECT - FILTERED BY PANCHAYAT) \*

Ward 01

Ward 02

Ward 05

S.No 120-180

+ Add more

Cancel

✓ Create User Account

### 3. Master Data Management & Analytics Dashboards

#### 3.1 Screen 3: A1 Master Data Forms (/admin/masters)

SUPER ADMIN (A1) • MASTER CONFIGURATION

Route: /admin/masters

1. Caste Master2. Job Master3. Party Master

+ Add New Caste Option

Enter Caste Name (e.g. BC - Nadar)

Status: Active (Yes)▼

+ Save Caste

| # | CASTE NAME        | STATUS | CREATED DATE | ACTIONS                |
|---|-------------------|--------|--------------|------------------------|
| 1 | BC - Nadar        | Active | 2026-08-10   | <div>EditDisable</div> |
| 2 | MBC - Thevar      | Active | 2026-08-10   | <div>EditDisable</div> |
| 3 | SC - Adi Dravidar | Active | 2026-08-11   | <div>EditDisable</div> |

#### 3.2 Screen 4: Panchayat-Wise Progress Dashboard

ANALYTICS & METRICS DASHBOARD

Route: /admin/dashboard & /subadmin/dashboard

TOTAL VOTERS

1,250,000

In Database

COMPLETED

850,000

68.0% Done

PENDING

400,000

32.0% Remaining

TODAY'S DONE

14,230

+12% vs Yesterday

| # | PANCHAYAT NAME | TOTAL VOTERS | UPDATED | PENDING | PROGRESS                  |
|---|----------------|--------------|---------|---------|---------------------------|
| 1 | Melur          | 25,000       | 20,000  | 5,000   | <div><div></div>80%</div> |
| 2 | Alanganallur   | 18,000       | 11,700  | 6,300   | <div><div></div>65%</div> |
| 3 | Othakadai      | 30,000       | 15,000  | 15,000  | <div><div></div>50%</div> |

## 4. Field Agent (A3) Mobile Survey Wireframe

Mobile-optimized field data collection form used by **Agent (A3)** during on-ground door-to-door survey verification.

09:41 AM

4G • 95%

Voter Field Survey

Agent: Ravi (#04)

SEARCH VOTER RECORD

TN/04/023/88991

Search

SYSTEM PRE-FILLED (READ-ONLY)

Locked

EPIC NUMBER

TN/04/023/88991

PANCHAYAT

Melur

DOOR NO / ADDRESS

12/4A, Gandhi St

AGE

34

SEX

Male

VOTER IDENTITY (CORRECTIONS ALLOWED)

VOTER FIRST NAME \*

Suresh

LAST NAME (HUSBAND / FATHER NAME) \*

M. Kumar

FIELD DATA TO BE FILLED BY AGENT

1. VOTER PHONE NUMBER \*

9845012345

2. CASTE (DROPDOWN FROM MASTER) \*

BC - Nadar

3. JOB / OCCUPATION (DROPDOWN FROM MASTER) \*

Private Business / Shop

4. BELONGS TO WHICH PART / PARTY (DROPDOWN) \*

Party A (DMK)

Clear

Save & Submit

## 5. Database Schema & REST API Specifications

### 5.1 PostgreSQL Table Definitions

```
-- 1. USERS & AUTH TABLE
CREATE TABLE users (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  mobile_number VARCHAR(10) UNIQUE NOT NULL,
  password_hash VARCHAR(255) NOT NULL,
  role VARCHAR(20) NOT NULL CHECK (role IN ('A1_SUPER_ADMIN', 'A2_ADMIN', 'A3_AGENT')),
  epic_id VARCHAR(50) UNIQUE NOT NULL,
  is_active BOOLEAN DEFAULT TRUE,
  created_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP
);

-- 2. USER JURISDICTION SCOPES (Cascading Permissions for A2 & A3)
CREATE TABLE user_jurisdictions (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  user_id UUID REFERENCES users(id) ON DELETE CASCADE,
  constituency_id INT NOT NULL,
  panchayat_id INT NOT NULL,
  ward_id INT NOT NULL
);

-- 3. A1 MASTER TABLES (Caste, Job, Party)
CREATE TABLE caste_master ( id SERIAL PRIMARY KEY, caste_name VARCHAR(100) UNIQUE NOT NULL, is_active BOOLEAN DEFAULT TRUE );
CREATE TABLE job_master ( id SERIAL PRIMARY KEY, job_title VARCHAR(100) UNIQUE NOT NULL, is_active BOOLEAN DEFAULT TRUE );
CREATE TABLE party_master ( id SERIAL PRIMARY KEY, party_name VARCHAR(100) UNIQUE NOT NULL, is_active BOOLEAN DEFAULT TRUE );

-- 4. VOTER SURVEY COLLECTION RECORDS
CREATE TABLE voter_surveys (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  epic_id VARCHAR(50) UNIQUE NOT NULL,
  corrected_first_name VARCHAR(100),
  corrected_last_name VARCHAR(100),
  phone_number VARCHAR(15) NOT NULL,
  caste_id INT REFERENCES caste_master(id),
  job_id INT REFERENCES job_master(id),
  party_id INT REFERENCES party_master(id),
  updated_by_agent_id UUID REFERENCES users(id),
  updated_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP
);
```

### 5.2 Core API Endpoints for Backend Developers

| METHOD | ENDPOINT ROUTE            | AUTHORIZED ROLE | DESCRIPTION & PAYLOAD                                                                 |
|--------|---------------------------|-----------------|---------------------------------------------------------------------------------------|
| POST   | /api/auth/login           | Public          | Takes { mobile_number, password } & returns JWT with role and assigned scope.         |
| GET    | /api/voters/verify-epic   | A1, A2          | Validates if EPIC ID exists in DB during registration. Returns citizen name and area. |
| POST   | /api/users/create         | A1, A2          | Creates user/agent with assigned constituency, panchayat, and ward list.              |
| GET    | /api/masters/dropdowns    | A1, A2, A3      | Returns active Castes, Jobs, and Party affiliations for dropdown inputs.              |
| GET    | /api/voters/search        | A3              | Searches voter by EPIC / Name / Door No within Agent's assigned scope.                |
| POST   | /api/voters/survey/submit | A3              | Saves/updates voter survey record and increments dashboard counters.                  |

| METHOD | ENDPOINT ROUTE       | AUTHORIZED<br>ROLE | DESCRIPTION & PAYLOAD                                                   |
|--------|----------------------|--------------------|-------------------------------------------------------------------------|
| GET    | /api/dashboard/stats | A1, A2             | Returns Panchayat-wise and Agent-wise progress counts and completion %. |